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Editor
Journal of Chemical Theory and Computation

Dear Editor,

Please consider the manuscript entitled “Non-Redundant and Symmetry-Adapted Coordinates for Black-Box Molecular Structure Refinement” for publication in the *Journal of Chemical Theory and Computation*.

The manuscript addresses a persistent limitation of semiexperimental molecular structure refinement: the fitted geometry often depends on a user-defined Z-matrix or on manually selected internal-coordinate models. This dependence is benign for simple acyclic molecules, but it becomes problematic for rings, fused systems, bridged frameworks, planar molecules, and cases with incomplete isotopic substitution data.

We introduce **SEfit**, a framework in which the structure is read as Cartesian coordinates and the active refinement space is generated automatically from molecular topology and symmetry. Two non-redundant, symmetry-adapted coordinate representations are available: generalized internal coordinates (GICs) built from homogeneous primitive blocks, and symmetry-adapted Cartesian displacements with translations and rotations removed. Chemical constraints, predicates, and parameter classes remain defined in primitive internal coordinates and are projected onto the chosen active basis, so the method preserves chemical interpretability without requiring a Z-matrix.

The work extends the semiexperimental structure-refinement capabilities developed in the **MSR** molecular-structure-refinement workflow. It retains moment-of-inertia fitting, vibrational and electronic corrections, parameter classes, reduced-dimensionality hydrogen models, covariance propagation, Kraitchman diagnostics, and human-readable reports, while moving coordinate construction and constraint handling to a fully automatic Cartesian/topology layer.

The approach is validated on glycolaldehyde, glycine II, cyclopentadiene, nitrobenzene, azulene, and norcamphor. These examples span standard acyclic, functional-constraint, cyclic, planar aromatic, fused-ring, and bridged cases. The results show that the symmetry-adapted GIC and Cartesian active spaces give consistent ranks, effective dimensionalities, and rotational residuals, while final bond lengths, angles, dihedrals, and their uncertainties are obtained from the optimized Cartesian structures with rigorous error propagation.

We believe the manuscript is appropriate for the *Journal of Chemical Theory and Computation* (JCTC) because it presents a general computational methodology for coordinate-dependent inverse problems, not only a molecule-specific refinement. The same coordinate layer is designed to support semiexperimental structures, Gaussian input generation, Wilson G/F matrix and potential-energy-distribution (PED) analysis, and related vibrational workflows.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The author has approved the submission and declares no competing financial interest. The software, benchmark inputs, generated tables, and regression tests can be made available to reviewers and, upon publication, under the distribution model selected by the author.

Sincerely,

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